

ELEC4631 Tutorial 8

1. Show that the following is true:

(a) For any real numbers x_1, x_2 , and $0 \leq \alpha \leq 1$,

$$(\alpha x_1 + (1 - \alpha)x_2)^2 \leq \alpha x_1^2 + (1 - \alpha)x_2^2.$$

(b) Let X be a symmetric positive definite $n \times n$ matrix. Prove that $X > 0$ if and only $X^{-1} > 0$.

(c) Let X be a symmetric positive definite $n \times n$ matrix. Prove that $X > 0$ if and only $WXW^T > 0$ for any invertible $n \times n$ matrix W (**note:** this claim also holds when positive definiteness (> 0) is relaxed to positive semidefiniteness (≥ 0)).

2. Using the abstract definition of a convex set, show that the following 2D sets are convex:

(a) The hexagon H described by

$$H = \{(x, y) \in \mathbb{R}^2 \mid x \geq -a, x \leq a, y - b - x \leq 0, y - b + x \leq 0, \\ y + b - x \geq 0, y + b + x \geq 0\},$$

with constants $b > a > 0$.

(b) The pizza slice sector P described by

$$P = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq r, y - 3x \leq 0, y - x \geq 0\},$$

with a constant $r > 0$.

3. Suppose that C_1 and C_2 are convex sets in $\mathbb{R}^n$. Show that $C_1 \cap C_2$ is either empty or convex. By induction then show that if $C_1, C_2, \dots, C_n$ are convex sets then $C_1 \cap C_2 \cap \dots \cap C_n$ is either empty or is a convex set for any $n \geq 2$.

4. Consider the strict LMI with decision variable $(x_1, x_2) \in \mathbb{R}^2$:

$$F(x) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + x_1 \begin{bmatrix} -3 & -1 \\ -1 & 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} > 0$$

- (a) Compute the characteristic equation

$$c_{F(x)}(\lambda) = \det(\lambda I - F(x)).$$

- (b) Derive the conditions on x_1, x_2 such that $F(x) > 0$ by finding conditions on x_1, x_2 such that all roots of $c_{F(x)}(\lambda)$ are positive.

- (c) Is the LMI feasible?

5. Let X be a symmetric 2×2 matrix, $X = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix}$. Consider the following pair of LMIs in X :

$$\begin{array}{rcl} X & > & 4I \\ \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} - X & < & 0 \end{array}$$

Express the above LMIs in the standard form with decision variable $x = (x_1, x_2, x_3)^T$.